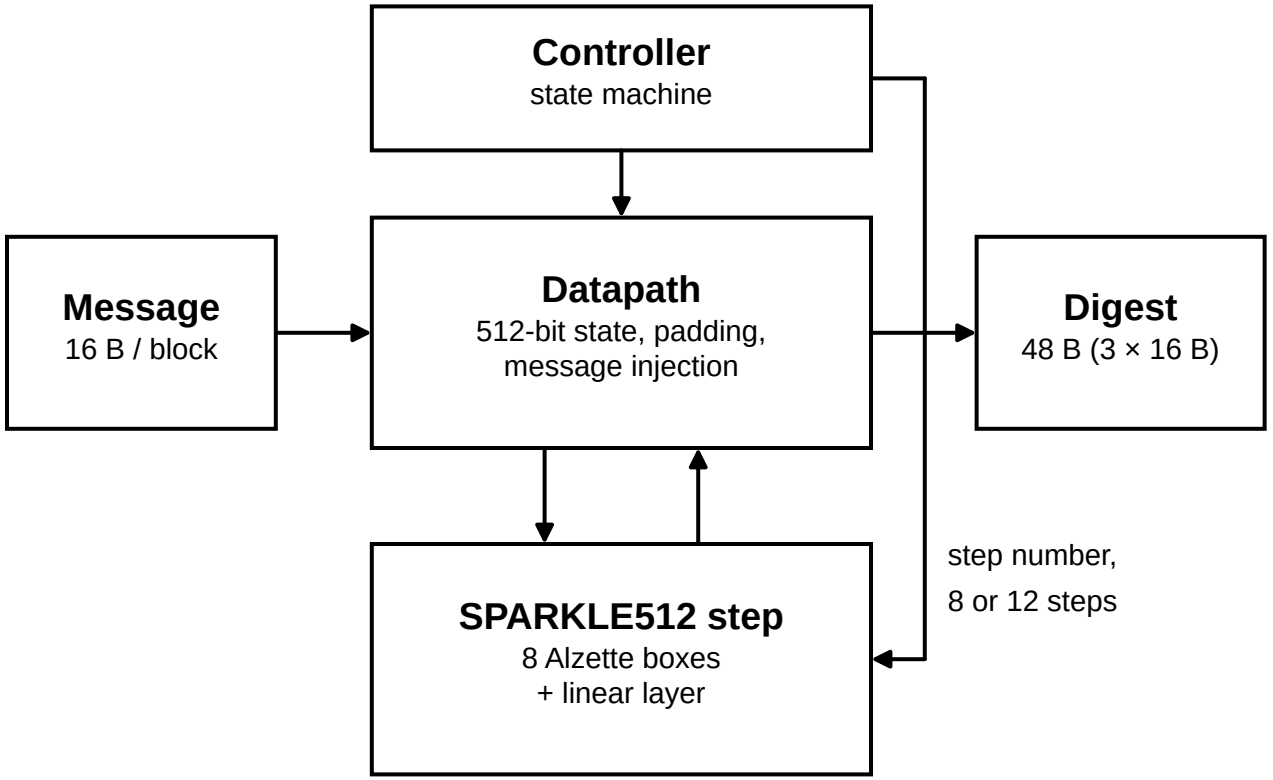
Esch384 Hash on FPGA: Progress Report

Henry Su, Nick Ngo | ECE 545 Final Project | Design Document

Design: iterative, one step per clock cycle



Arty A7 (Artix-7), Vivado 2026.1, Claude Code (AI, all use logged)

Goal: smallest area

Serialize: 8 → 4 → 2 → 1 Alzette boxes

Compare LUTs, flip-flops, clock speed, throughput

Progress (Henry: done; Nick: next)

Done: reference model, all 1025 official tests

Done: Alzette, linear layer, SPARKLE512

all tests pass; 2023 LUTs, 521 FFs, 76 MHz

Next: datapath, controller, full hash (Nick)

Risks and challenges

Each block tested alone against official C code

Unclear byte order → checked with C code

Plan: Nov 17 datapath + controller → Nov 21 full hash → Nov 24 synthesis → Dec 1 area optimization → Dec 9 final presentation